

Feature Scaling (Part 1)

This technique helps the gradient descent to run much faster.

Feature & params value:

$$\widehat{\text{price}} = w_1 x_1 + w_2 x_2 + b$$

$\downarrow$ size $\downarrow$ no. of bedrooms

relatively large values
 $\downarrow$

x_1 : size (ft²)

range: 300-2000

x_2 : # bedrooms

range: 0-5

$\uparrow$
relatively small values

lets take example of house:

$$x_1 = 2000, x_2 = 5, \text{price} = \$500k$$

↳ for this what are reasonable values for

w_1, w_2 ?

$$\text{Say } w_1 = 50, w_2 = 0.1, b = 50$$

in this case price will be:

$$\begin{aligned}\widehat{\text{price}} &= 50 \times 2000 + 0.1 \times 5 + 50 \\ &= 100,000k + 0.5k + 50k\end{aligned}$$

$$\widehat{\text{price}} = \$100,050.5k$$

↳ clearly v far from actual \$500k.

Not a v good set of parameter choices.

Another possibility:

Now,

$$w_1 = 0.1, w_2 = 50, b = 50$$

$$\begin{aligned}\widehat{\text{price}} &= 0.1 \times 2000 + 50 \times 5 + 50 \\ &= 200k + 250k + 50k \\ &= \$500k\end{aligned}$$

Much more
reasonable

- We noticed that when a possible range of values of a feature is large like $x_1(300-2000)$ [size (ft²)]:

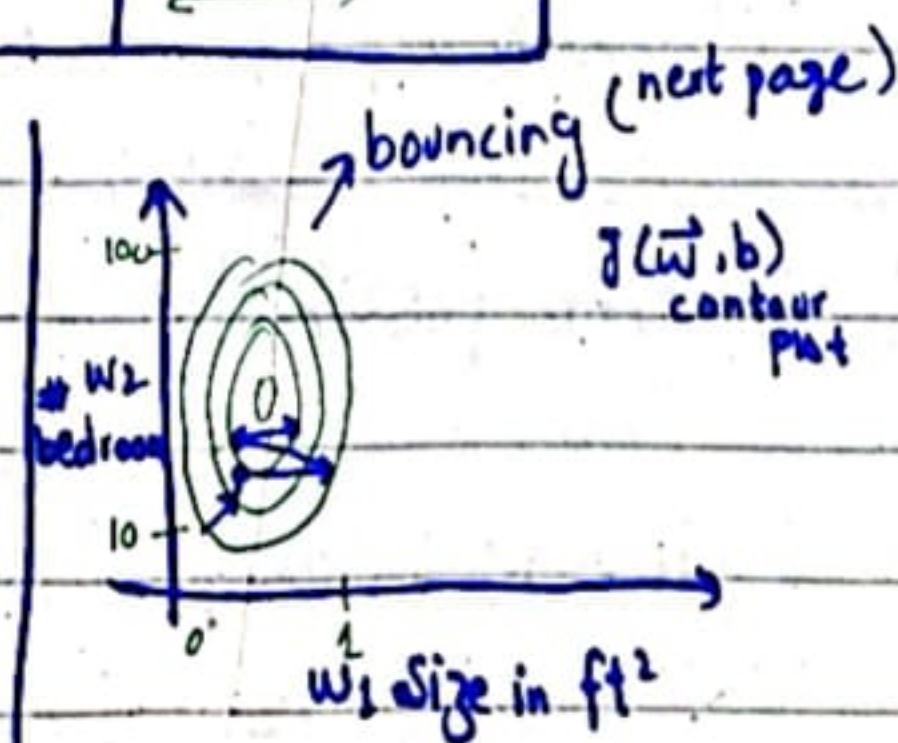
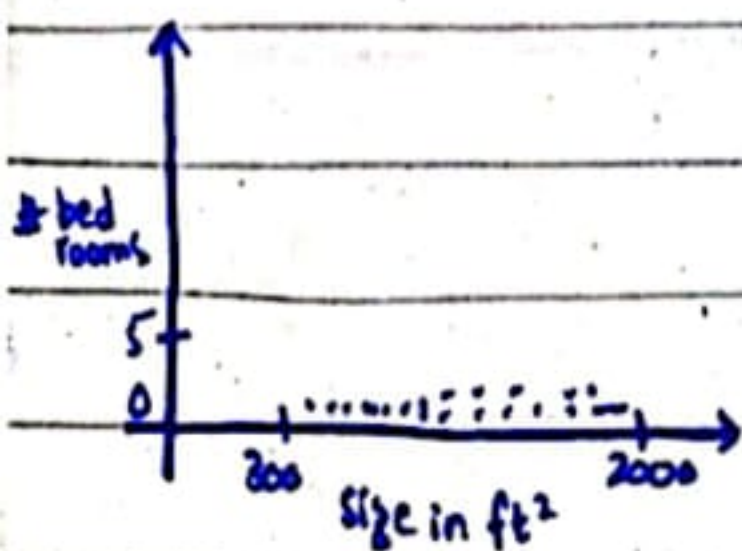
It's more likely that a good model will choose a relatively small param value like 0.1.

&

- when possible range of values of a feature is small like $x_2(0-5)$ [# of bedrooms], a good model will choose a relatively large value like 50.

How does this relate to gradient descent?

	size of feature x_i	size of param w_i
size in ft ²	←→	←→
#bedroom	←→	←→



• You might see a contour plot where the horizontal axis has much narrower range (0-1) whereas vertical has like (10-1000).

• so contour form oval or ellipses that are short on one side & large on other.

↳ this is because a very small change in w_1 can have very large impact on estimated price and that's very large impact on J , as it tends to be multiplied by a very large number (the size in sq.ft).

In contrast,

takes much larger change in w_2 in order to change predictions much, and thus small changes to w_2 don't change cost function nearly as much.

→ as in previous page gradient descent may bounce back & forth as contours are skinny. before it can reach global minimum.

What's the solution then?

In situation like these, useful thing to do is scale the features (meaning transformation of your training data so that x_1 may range from 0-1 & x_2 also 0-1).

